



Chapter 12 — Appendices

Appendix A — Repository Structure

```
doip/
├─ frontend/                # React + TypeScript + Vite
│   └─ src/
│       └─ app/             # routing, providers, layout
shell
│   │   └─ features/
│   │   │   └─ map/         # MapLibre layers, draw tool
s, replay
│   │   │   └─ dashboard/   # widgets grid
│   │   │   └─ missions/    # planner UI, optimization v
iews
│   │   │   └─ ai/          # sitrep timeline, chat, RAG
panel
│   │   │   └─ vision/      # job submit, filmstrip revi
ewer
│   │   │   └─ logistics/   # inventory, forecasts
│   │   │   └─ personnel/   # vitals roster
│   │   │   └─ analytics/   # after-action reports
│   │   └─ lib/ws/          # WebSocket client, resume l
ogic, reducers
│   │   └─ lib/api/         # typed API client (generate
d from OpenAPI)
│   └─ tests/              # vitest + Playwright
└─ backend/               # FastAPI modular monolith
    (gateway + domains)
    │   └─ app/
    │   │   └─ api/         # routers (v1), WS gateway
    │   │   └─ core/        # config, auth, RBAC, rate l
imiting, logging
    │   └─ events/         # event schema registry, emi
```

```

tter, bus adapters
|   |   | — missions/
tra, feasibility
|   |   | — map/
ries, GeoJSON
|   |   | — logistics/
lds
|   |   | — personnel/
|   |   | — db/
migrations
|   | — tests/
n-run
| — simulation/
backend, runnable solo)
|   | — engine/
tems, emitter
|   | — scenarios/
ON schema
|   | — dsl/
(whitelisted)
| — ai-service/
|   | — pipelines/
outer
|   | — providers/
rs
|   | — eval/
|   | — corpus/
arkdown)
| — vision-service/
kers
| — analytics/
e pipeline, registry
| — shared/
tracts (events, DTOs)
| — docker/
e.yml, nginx conf
| — kubernetes/
(Phase 7)
# domain: planning, A*/Dijks
# domain: zones, spatial que
# domain: resources, thresho
# domain: vitals projections
# SQLAlchemy models, Alembic
# unit + integration + golde
# the engine (importable by
# clock, world state, subsys
# YAML scenario library + JS
# condition/action functions
# LLM + LangGraph + RAG
# sitrep graph, rag graph, r
# ollama / hosted-API adapte
# golden sets + eval harness
# authored SOPs & manuals (m
# YOLO + OpenCV + Celery wor
# forecasting models, featur
# cross-service Pydantic con
# Dockerfiles, docker-compos
# Helm chart, values files

```

```

├─ docs/                                # ADRs, runbooks, this specification, demo scripts
├─ .github/workflows/                  # CI/CD pipelines
├─ Makefile                            # dev / test / seed / eval-a
└─ i / demo targets

```

Appendix B — Glossary

Term	Definition
Run	One execution of a scenario by the simulation engine; the top-level container for all operational data
Scenario	A YAML-authored script defining entities, timeline events, and reactive rules for a run
Tick	One step of the simulation clock (1 sim-second at speed 1x)
Sim time / wall time	Simulated clock vs real-world clock; speed multiplier links them
Event	An immutable, typed record of something that happened; the atomic unit of the platform
Projection	Mutable current-state tables derived from the event stream (units, incidents, ...)
Sitrep	Situation report — an AI-generated, event-cited summary of recent developments
RAG	Retrieval-Augmented Generation — LLM answers grounded in retrieved document chunks
Grounding verification	Post-generation check that every AI claim maps to real source events/chunks
Golden run	A seeded reference simulation whose exact event stream is a committed test fixture
COP	Common Operational Picture — the single fused view of all operational information
Sector	A named polygon subdividing the operational area, used for aggregation and reporting
Restricted zone	A polygon that routes must never cross (hard planning constraint)
HPA	Horizontal Pod Autoscaler — scales vision workers on queue depth in Kubernetes
ADR	Architecture Decision Record — a short document capturing a decision and its rationale

Appendix C — Learning Resources per Skill Area

Area	What to study	Applied in
GIS fundamentals	PostGIS docs (<code>ST_DWithin</code> , <code>ST_Intersects</code> , GiST indexes); GeoJSON spec (RFC 7946)	Ch. 5 queries, map layers
MapLibre	Official examples: sources/layers, symbol animation, draw plugins	Live map, replay
Async Python	<code>asyncio</code> docs; FastAPI WebSocket guide	Engine loop, WS gateway
Event-driven design	“Designing Data-Intensive Applications” (Kleppmann), ch. 11	Event schema, bus, replay
Pathfinding	Red Blob Games A* interactive guides	Mission planner
Local LLMs	Ollama docs; LangGraph tutorials	AI service
RAG quality	pgvector README; hybrid search patterns; RAG evaluation guides	Knowledge assistant
Object detection	Ultralytics YOLO docs; VisDrone / xView dataset papers	Vision service
Forecasting	Prophet quickstart; XGBoost docs; quantile regression	Analytics
Kafka	Confluent free courses — topics, partitions, consumer groups	Phase 7 migration
Kubernetes	k3s docs; Helm chart tutorial	Phase 7 deploy
Observability	Prometheus + Grafana getting-started; RED/USE method	Ch. 10 monitoring

Appendix D — Frequently Asked Questions

Q: Is this a real military system?

No. DOIP is a simulation, training, and planning platform. Every unit, sensor, and incident is synthetic. It controls nothing physical and connects to no real-world operational systems (see Ch. 1 §1.8).

Q: Why the defense framing if scenarios are disaster-response?

The *architecture pattern* (common operational picture, sitreps, logistics, mission planning) originates in command-and-control design. The *content* is deliberately civilian: floods, fires, search-and-rescue. This maximizes engineering realism while staying ethically and legally clean.

Q: Can this run without a GPU?

Yes. Quantized 7B LLMs on CPU, YOLO nano at 2 fps sampling, and all latency budgets in Ch. 8 are sized for CPU-only hardware (Ch. 4 §4.6).

Q: What does it cost to run?

Nothing. Every component is open-source and self-hosted. Optional: a cheap VPS for a public demo.

Q: Why not start with Kafka and Kubernetes since they're in the spec?

Because a working product must exist before its scale story. Redis Pub/Sub and Docker Compose deliver the identical demo experience in Phases 1–6; the Phase-7 migrations then demonstrate *evolution* — a rarer and more valuable skill than greenfield setup (ADR-2, Ch. 3).

Q: How is this different from a Grafana dashboard over fake data?

Three things: (1) a deterministic, scriptable simulation engine with a scenario DSL; (2) a geospatial planning layer with real algorithms and constraints; (3) grounded AI with verifiable citations. Each is a system, not a widget.

Q: What's the single most important thing to build well?

The simulation engine (Ch. 7). Every other module is a consumer of its events. If the engine is deterministic, observable, and well-tested, the rest of the platform assembles on solid ground.

Q: How do I extend DOIP after Phase 7?

Natural extensions: multi-operator collaboration (shared cursors, role-based consoles), scenario marketplace/editor UI, reinforcement-learning agents as simulated adversaries/teammates, 3D terrain (MapLibre terrain or [deck.gl](#)), and voice-driven operator assistant.

Appendix E — Document Control

Field	Value
Version	1.7
Status	Approved baseline for Phase 0 kickoff
Author	Saranraj U
Created	21 July 2026
Review cadence	End of every phase — update affected chapters + changelog below

Changelog

Version	Date	Changes
1.0	2026-07-21	Initial complete specification: all 12 chapters
1.1	2026-07-21	Added Chapter 13 — AI-Generated Scenario Simulation & Animated Playback ("What-If Theater")
1.2	2026-07-21	Added Chapter 14 — Budget, Capability & Historical Benchmarking Module (Preparedness Investment Advisor)
1.3	2026-07-21	Added Chapter 15 — Multi-Operator Collaboration & War-Room Mode and Chapter 16 — Adversarial Red-Team Simulation Agent; updated Chapter 3 system architecture (§3.10 extension components)
1.4	2026-07-21	Added Chapter 17 — Project & Platform Workflows
1.5	2026-07-21	Added Chapter 18 — UI/UX Design Specification and Chapter 19 — Starter Content, Licensing & Zero-Cost Stack; updated Chapter 4 with YOLO licensing note and zero-cost cross-reference
1.6	2026-07-21	Refocused platform framing on defense & military use: Chapter 1 positioning, Chapter 19 scenario library rebalanced to military operations (HADR retained as a military mission and benchmarking data source)
1.7	2026-07-28	Chapter 18: added ASCII wireframes for all 15 screens (§18.6), replacing the partial 3-screen set